

7

Fans, Arrangements, Zonotopes, and Tilings

Zonotopes are the images of n -cubes under affine projection maps. Since for most aspects of polytope theory n -cubes are not very complicated, this definition may hide the complexity and richness of this concept. Zonotopes are interesting from various points of view. Their combinatorial structure is closely linked to (and in a precise sense equivalent to) that of real linear hyperplane arrangements.

The aim of this lecture is to provide basic geometric intuition and the tools for a combinatorial description of zonotopes. We will see how zonotopes again are modeled by oriented matroids, and discuss the surprising appearance of general (“nonrealizable”) oriented matroids in the study of zonotopal tilings, and of hyperplane arrangements.

7.1 Fans

Definition 7.1. A *fan* in $\mathbb{R}^d$ is a family

$$\mathcal{F} = \{C_1, C_2, \dots, C_N\}$$

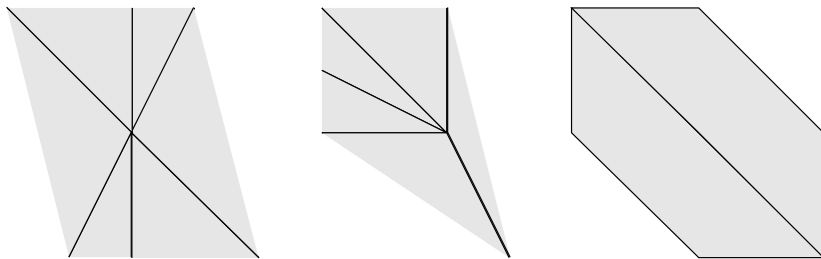
of nonempty polyhedral cones, with the following two properties:

- (i) Every nonempty face of a cone in $\mathcal{F}$ is also a cone in $\mathcal{F}$.
- (ii) The intersection of any two cones in $\mathcal{F}$ is a face of both.

The fan $\mathcal{F}$ is *complete* if the *union* $\bigcup \mathcal{F} := C_1 \cup \dots \cup C_N$ of $\mathcal{F}$ is $\mathbb{R}^d$, that is, if $\bigcup_{i=1}^N C_i = \mathbb{R}^d$. We will consider only complete fans here, and thus we will omit the word “complete” most of the time.

$\mathcal{F}$ is *pointed* if $\{\mathbf{0}\}$ is a cone in $\mathcal{F}$ (and thus is a face of every cone in $\mathcal{F}$). It is *simplicial* if all its cones are simplicial cones, that is, cones spanned by linearly independent vectors. Simplicial cones and fans are automatically pointed.

The following figure shows three complete fans in $\mathbb{R}^2$, with $N = 13$, $N = 11$, and $N = 3$ cones, of which 6, 5, and 2, respectively, are full-dimensional. The first two fans are pointed (for $d = 2$ this implies they are simplicial); the third one is not.



There are various equivalent or similar ways to define fans (see also the notes to this lecture). What we have given here essentially just describes a polyhedral complex of cones (as in Definition 5.1) whose union is $\mathbb{R}^d$. In particular, the definition implies that the relative interiors of the cones in $\mathcal{F}$ form a partition of space:

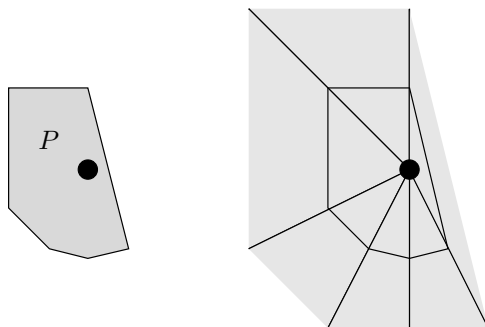
$$\bigcup_{i=1}^N \text{relint}(C_i) = \mathbb{R}^d.$$

Now comes the first reason why we look at fans in the theory of polytopes.

Example 7.2. Let P be a polytope in $\mathbb{R}^d$ with $\mathbf{0} \in \text{relint}(P)$. We define the *face fan* of P as the set of all the cones spanned by proper faces of P :

$$\mathcal{F}(P) := \{\text{cone}(F) : F \in L(P) \setminus \{P\}\}.$$

$\mathcal{F}(P)$ is a pointed fan in $\text{lin}(P)$: its union is the linear hull $\text{lin}(P)$. It is a complete fan in $\mathbb{R}^d$ if P is a d -polytope, with $\mathbf{0} \in \text{int}(P)$.



Our figure indicates the construction of the face fan, for a 2-polytope with a given origin in it. Note that the geometry of the face fan does depend on the position of the origin in P .

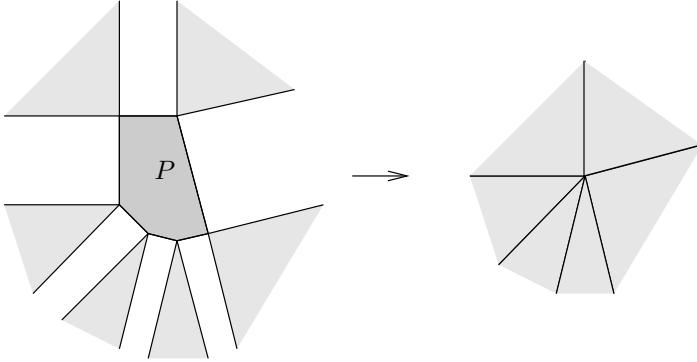
Example 7.3. Let P be a nonempty polytope in $\mathbb{R}^d$. For the *normal fan* of P we take the cones of those linear functions which are maximal on a fixed face of P . That is, for every nonempty face F of P we define

$$N_F := \left\{ \mathbf{c} \in (\mathbb{R}^d)^* : F \subseteq \{ \mathbf{x} \in P : \mathbf{c}\mathbf{x} = \max_{\mathbf{y} \in P} \mathbf{c}\mathbf{y} \} \right\},$$

and with this we define

$$\mathcal{N}(P) := \{ N_F : F \in L(P) \setminus \{\emptyset\} \}.$$

$\mathcal{N}(P)$ is a complete fan in $(\mathbb{R}^d)^*$. If P is d -dimensional, then the fan is pointed, since then $\{\mathbf{0}\} = N_P$ is in the fan.



The above figure illustrates the construction of the normal fan of a 2-polytope: for this we have identified $\mathbb{R}^2$ with $(\mathbb{R}^2)^*$ via the usual scalar product, which accounts for the right angles in the figure.

The face fan and the normal fans are very natural objects associated with a polytope. In particular, they come up in the theory of optimization. For this note that the question “Which cone of $\mathcal{N}(P)$ does $\mathbf{c}$ lie in?” is the *linear programming problem* $\max \mathbf{c}\mathbf{x} : \mathbf{x} \in P$. Similarly, the question “Which cone of $\mathcal{F}(P)$ does $\mathbf{v}$ lie in?” is the *separation problem* of finding one single valid inequality that determines whether $\alpha\mathbf{v} \in P$, for various $\alpha > 0$.

Example 7.4. Let $\mathcal{A} := \{H_1, \dots, H_p\}$ be a finite set of linear hyperplanes in $\mathbb{R}^d$, where each H_i is of the form $H_i = \{\mathbf{x} \in \mathbb{R}^d : \mathbf{c}_i\mathbf{x} = 0\}$ for some $\mathbf{c}_i \in (\mathbb{R}^d)^*$.

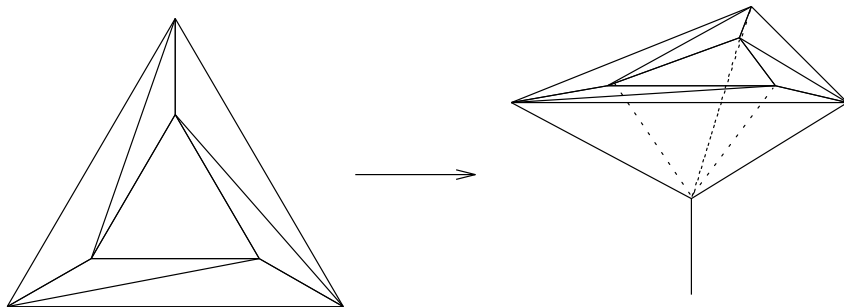
Clearly, the arrangement $\mathcal{A}$ decomposes $\mathbb{R}^d$ into a complete fan $\mathcal{F}_{\mathcal{A}}$. The cones of the fan are also referred to as the *faces* of the (linear) hyperplane

arrangement $\mathcal{A}$. The combinatorics of the fan encodes a lot about the configuration of row vectors $C = \{c_1, \dots, c_p\}$ — it is quite easy to see that the cones in $\mathcal{F}$ are in natural correspondence with the covectors of C . We'll see details about this in Theorem 7.16 and Corollary 7.18.

In the figure after Definition 7.1, the first and the third fan are given by hyperplane arrangements, the second one is not.

Example 7.5. There are also complete simplicial fans $\mathcal{F}_0$ in $\mathbb{R}^3$ that are *nonpolytopal*, that is, not of the form $\mathcal{F}(P)$ for any polytope P .

For one possible construction, we start from a simplicial 2-diagram $\mathcal{D}$ that is not a Schlegel diagram, for example the one we constructed before Theorem 5.7. This we place into an affine plane, and take all the cones over faces of the diagram. We complete $\mathcal{F}_0$ by adding one extra ray and the simplicial cones that are spanned by this ray together with the cones generated by the boundary of the diagram $\mathcal{D}$. Our figure illustrates the construction:



This fan $\mathcal{F}_0$ is not polytopal. In fact, assume that P is a polytope with $\mathcal{F}_0 = \mathcal{F}(P)$. Now consider the polytope P' given by the convex hull of all vertices of P except for the one on the “extra ray.” Then the origin is beyond a triangular facet of P' , and the corresponding Schlegel diagram would be (projectively) equivalent to the one we started with. This yields a contradiction.

This example shows by far not the worst that can happen: for example, there exists a (nonsimplicial) fan that is not even the face fan of a *starshaped sphere* (with flat facets); see Eikelberg [194]. We also refer to Ewald [201, Sect. III.5] and the examples and references he gives.

We continue with three more “trivial” operations on fans, which will turn out to be very valuable soon.

Definition 7.6. If $\mathcal{F}$ is a fan in $\mathbb{R}^p$, and $\mathcal{G}$ is a fan in $\mathbb{R}^q$, then their *direct sum* is the fan

$$\mathcal{F} \oplus \mathcal{G} := \{C \times C' : C \in \mathcal{F}, C' \in \mathcal{G}\}.$$

If $\mathcal{F}$ and $\mathcal{G}$ are both fans in the same space $\mathbb{R}^d$, then we define their *common refinement* as

$$\mathcal{F} \wedge \mathcal{G} := \{C \cap C' : C \in \mathcal{F}, C' \in \mathcal{G}\}.$$

If $\mathcal{F}$ is a fan in $\mathbb{R}^d$ and $V \subseteq \mathbb{R}^d$ is a vector subspace, then the *restriction* of $\mathcal{F}$ to V is the fan

$$\mathcal{F}|_V := \{C \cap V : C \in \mathcal{F}\}.$$

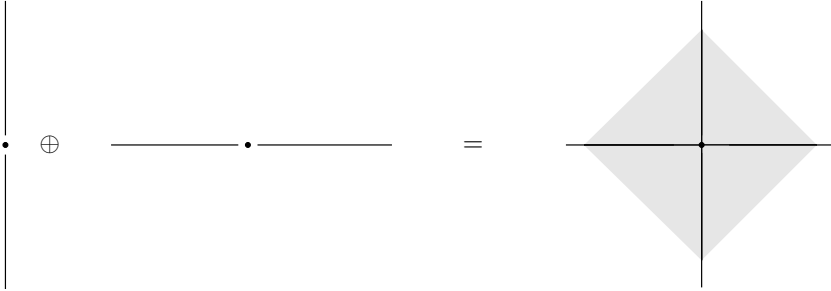
It is easy to check that all three constructions are again fans in the sense of Definition 7.1. If you want, you can consider $\mathcal{G}_V := \{V\}$ as a (noncomplete) fan in $\mathbb{R}^d$: then the restriction to V is the intersection with $\mathcal{G}_V$, that is, $\mathcal{F}|_V = \mathcal{F} \wedge \mathcal{G}_V$.

Lemma 7.7. *Let $P \subseteq \mathbb{R}^p$ and $Q \subseteq \mathbb{R}^q$ be two polytopes. Then the normal fan of the product $P \times Q \subseteq \mathbb{R}^{p+q}$ is the direct sum*

$$\mathcal{N}(P \times Q) = \mathcal{N}(P) \oplus \mathcal{N}(Q). \quad \square$$

Example 7.8. The normal fan of the cube $C_d = \{\mathbf{x} \in \mathbb{R}^d : -1 \leq x_i \leq 1\}$ coincides with the face fan of its polar, the d -crosspolytope (cf. Exercise 7.1 for the general case of this).

We find that the normal fan is given by the arrangement of coordinate hyperplanes in $(\mathbb{R}^d)^*$. This is quite trivial (since it is trivial to optimize over a cube), and it also confirms Lemma 7.7: the fan of the arrangement of coordinate hyperplanes is the direct sum of 1-dimensional fans.



Also, the cones in the fan are characterized by the sign function: each cone in the fan can be identified with a vector in $\{+, -, 0\}^d$, and the orthants correspond to the sign vectors in $\{+, -\}^d$.

7.2 Projections and Minkowski Sums

Let $P \subseteq \mathbb{R}^p$ be a p -polytope, and let $\pi : \mathbb{R}^p \longrightarrow \mathbb{R}^d$ be an *affine map*,

$$\pi(\mathbf{x}) = A\mathbf{x} - \mathbf{z},$$